

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: VII

Subject Code & Name: BTCVC 703 Construction Techniques

Max Marks: 60

Date: 06/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is ripper? What are the different types of rippers? State the applications of Rippers	CO1	6
B) Enlist the various types of Rollers and explain any one.	CO1	6
C) Write down working and application of power shovel with neat labelled diagram.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Enlist various methods of dewatering. Explain different method of dewatering in excavation.	CO2	6
B) What are the different properties which are considered in selecting an explosives?	CO2	6
C) Define jack hammers. Explain the working of jack hammer.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Describe the process of underwater concreting with neat labelled diagram.	CO3	6
B) Define grouting. Describe the various materials used for grouting. Write down advantages of grouting.	CO3	6
C) Explain the working of ready-mix concrete plant with their components. Draw a neat labelled diagram.	CO2	6
Q.4 Solve Any Two of the following.		12
A) What do you mean by Hoisting Equipment? Explain Rail mounted tower crane with suitable diagram.	CO4	6

B)	Enlist various types of dredgers. Explain with neat labelled diagram: Grab dredger	CO4	6
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C)	Write down advantages and disadvantages of pre-fabrication.	CO4	6
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Q. 5	Solve Any Two of the following.		12
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A)	Write down application of Diaphragm wall. Explain the method of construction of Diaphragm wall.	CO3	6
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B)	Explain the operation involved in hot mix plan with neat sketch.	CO4	6
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C)	What are the causes of accidents on construction site? How to prevent it.	CO4	6
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